Exercises with Functions

These exercises focus on creating and using our own functions:

- ⋆ def - defining a function
- ⋆ parameters
- ⋆ return values
- ⋆ calling a function

Create Python programs to do the following:

1. Write a function called **CtoF** that takes a temperature in Celsius and returns the equivalent temperature in Fahrenheit.

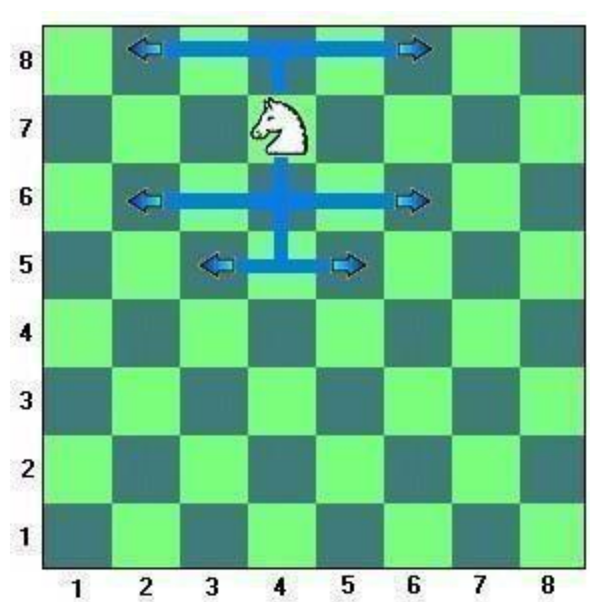
2. Write a function called **letterMark** that takes an integer mark out of 100 and returns the proper letter grade (a string).
   e.g. **letterMark**(76) → B

3. Write a function called **nickname** that creates a name from two given strings. The function takes 4 parameters (first 2 are the names, followed by 2 numbers). e.g. nickname("Sheldon","Amy",2,3) will return the value "Shamy" (make sure it is not "ShAmy")

4. A palindrome is a word that reads the same both forward and backwards. Create a function called **palindrome** that takes a single string as a parameter and returns True if the word is a palindrome and False otherwise. (Ignore punctuation, capital and lowercase letters)

   (e.g. "Madam, I'm Adam" is a palindrome)

5. Knights in chess move in an L shape. They go 2 in one direction then one space at a 90 angle, as the diagram shows. Create a function called **knightMoves** that takes in the current location (row, col) of a knight on the chess board and returns a list of tuples (row, col) of all possible places the knight can move.



6. Write a function called **drawface** that draws a happy face of the size and at the location specified by the parameters. It is important that the features of the face are always in the same proportion.

7. Write a function called **perColour** that will take a colour tuple and return the percentage of the screen that is that colour. When you test your function make sure that you draw something on the screen first. (hint: use screen.get_at())